Determinants of Incident Prediabetes and Type 2 Diabetes in a 7-year Cohort in a Developing Country: The Isfahan Cohort Study

Running title: Determinants of incident prediabetes and type 2 diabetes

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Abstract

Background: This study aimed to identify determinants of new onset type-2 diabetes (T2D) and prediabetes in an Eastern Mediterranean developing community, a region with high rate of T2D but scarcity of evidence.

Methods: Participants free of T2D were identified in a population-based study of adults older than 35 years living in the rural and urban areas of three districts in Iran in 2001 and were re-examined after 7 years. Biochemical measurements, anthropometric data, blood pressure, smoking status and dietary intake were determined at both measurement points through identical protocols and procedures. Data analysis was done using multivariate ordered logistic regressions.

Results: Incidence rate of prediabetes and T2D was 32.3(95%CI, 29.7–35.1) and 18.9(17.1–20.9) per 1000 person-year, respectively. Odds of progression to prediabetes and T2D were increased with ageing, living in rural area (OR=1.28), unhealthy diet (OR=1.32), overweight (OR=1.45), obesity (OR=1.97), waist gain (OR=1.37), high waist-to-hip ratio (OR=1.35), hypertension (OR=1.35), and hypertriglyceridemia (OR=1.50); but reduced with higher education (OR=0.76) and weight loss (OR=0.67) in adjusted model. In those with prediabetes at baseline, parental history of T2D (OR=2.26), obesity (OR=2.59), high waist circumference (OR=2.03), and hypertriglyceridemia (OR=1.94) were associated with the increased odds of developing T2D. No factor was found to be associated with regression from prediabetes to normoglycemia in adjusted models.

Conclusions: Current data from a region with the highest expected prevalence of T2D clearly endorses proposed statistical projections for future trend of diabetes, and highlights potential role of obesity specifically visceral type and hypertension as associated modifiable risk factors.

KEYWORDS: Type 2 Diabetes, Prediabetes, Risk Factor, Obesity

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Significant findings of the study: Progression to prediabetes and diabetes were associated with rural area, unhealthy diet, obesity, waist gain, hypertension, and hypertriglyceridemia; but were prevented by education and weight loss. In prediabetes, parental history, obesity, and hypertriglyceridemia increased odds of developing diabetes.

What this study adds: Incidence of prediabetes was even higher than diabetes, but a considerable proportion were normalized. Several factors were involved in progression to diabetes but hypertriglyceridemia, general and central obesity were associated in both normoglycemia and prediabetes. Weight loss and waist gain >10% also had impact.

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INTRODUCTION

Hyperglycaemia and diabetes are leading causes of mortality and morbidity worldwide, through micro-vascular and macro-vascular complications.[1] Remarkable increased risk of all-cause mortality and cardiovascular mortality were reported to be associated with new onset type 2 diabetes,[2] henceforth diabetes. By 2030, number of patients with diabetes in developing countries living in urban areas is projected to be doubled. By region, Middle Eastern Crescent is expected to have the highest percentage (163%) of change in number of people with diabetes within 2000-2030.[3] These estimations are very well correlated with recently published systemic analysis for trends in glycaemia and diabetes which has indicated upward trends of mean fasting plasma glucose (FPG) and diabetes globally, with highest reports from Oceania, North Africa, the Middle East and the Caribbean.[4]

Onset of diabetes is estimated to be much earlier in developing countries in comparison with developed ones, and higher number of patients with diabetes in 45- to 64-year-old age group is expected in developing communities for 2030 compared to the older age groups.[3] In fact, earlier onset of diabetes in the more economically productive years of life leads to an extra burden of cost of illness in these communities. National prevalence of diabetes in Iran has reported to be 8.7% among Iranians within the 25- to 64-year-old age group in 2007,[5] and more than 1.8 fold increase has reported for prevalence of diabetes in a period of 8 years since 1999.[6] Additionally, data from an urban cohort in Iran have shown that more than 1% of the Iranian urban population older than 20 years develop diabetes annually.[7] With respect to almost three times higher costs in patients with diabetes compared to those without diabetes in Iran,[8] diabetes has already started to be an alarming public health burden in Iran in different aspects.

Few longitudinal data regarding glucose intolerance (prediabetes and diabetes) and associated risk factors is at hand in Iran and Middle East and there are some limitations in terms of generalizability of available published data. Such information could be exploited by both clinicians and health care policy

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makers in the senior regulatory and official levels for long term public health strategies including public health prevention interventions. In this respect, Isfahan Cohort Study (ICS) was designed and implemented to provide an updated quantification of the growing public health burden of cardiovascular disease (CVD) and its risk factors such as diabetes with more generalizable data from an Eastern Mediterranean developing community. This report aimed to find factors associated with incident diabetes and prediabetes.

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METHODS:

The ICS is an ongoing prospective population-based study of adults living in rural and urban areas from three different districts (i.e. Isfahan, Arak and Najafabad) in central Iran. Subjects were participants of the baseline survey of Isfahan Healthy Heart Program (IHHP), a community for prevention and control of CVDs. Recruitment period started from January 2nd and lasted up to September 28th, 2001. Multistage random sampling was implemented for subject enrolment in order to reflect age, sex, and urban/rural distribution of the community. Sampling methodology and population characteristics have been described previously.[9]

Subjects with FPG more than 126 mg/dl (7.0 mmol/l) or the 2-hour post challenge glucose value more than 200 mg/dl (11.1 mmol/l) or being on antidiabetes medication were defined to have diabetes (which will be referred as "diabetes" onward). Prediabetes was defined if the 2-hour post load glucose was ≥ 140 mg/dl (7.8 mmol/l) and < 200 mg/dl (11.1 mmol/l) (IGT) and/or if the FPG was ≥ 100 mg/dl (5.5 mmol/l) and < 126 mg/dl (7.0 mmol/l) (IFG).[10]

Clinical and biochemical assessments

Survey methodology applied for baseline (2001) and follow-up (2007) examinations were almost identical. Trained health care professionals conducted medical interviews and physical examinations. To assess the routine dietary regimen of participants, a validated 48-item food frequency questionnaire (FFQ) was used.[11] Subjects were divided into quintiles after quantifying intakes of twelve food categories. Individuals located in first three quintiles were scored "0" and those in the two highest quintiles were scored "1" in healthy food groups, while vice versa was done for those in unhealthy food groups. The dietary score (0-12) was then calculated by summing up the scores for all food groups as an indicator for overall diet quality with higher scores demonstrating healthier diet.[12]

All the measurements of blood pressure (BP) and anthropometric parameters [weight, height, waist circumference (WC) and hip circumference] were conducted using calibrated instruments in the

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fashion that has been described previously.[9] Individuals who smoked daily were considered as smokers. Hypertension was defined as systolic BP ≥140 mmHg, and/or diastolic BP ≥90 mmHg or if individual was taking antihypertensive drugs. Overweight and obese individuals were those with body mass index (BMI) 25-29.9 kg/m^2 and ≥30 kg/m^2, respectively. To define metabolic syndrome (MetS), the harmonized definition was used and the modified cut-off points of WC for Middle Easterners was applied.[13] Blood sample was taken after a 10-hr fasting period to assay blood lipids (total cholesterol, triglyceride and HDL cholesterol).

Laboratory methods

Serum lipids and plasma glucose were analysed by enzymatic methods [Eppendorf, Hamburg, Germany (2001) and Hitachi 902, Japan (2007)] within 4 hours of specimen collection, respectively. LDL cholesterol was calculated using the Friedewald's equation.

Statistical analysis

Multivariate ordered logistic regression models were employed to examine the association of various factors with 7-year incidence of an ordinal prediabetes/diabetes outcome adjusted for age, sex, parental history of diabetes, and BMI. Proportional odds assumption was checked by Brant test. If relationship between each pair of outcome groups was not the same (P-value of Brant test <0.05), generalized ordinal logistic regression was also performed reporting separated OR for prediabetes and diabetes. Considering those who sustained prediabetes status as the reference level, factors significantly related to either development of diabetes or retrieve of normoglycaemia were evaluated using generalized ordinal logistic regression analysis.

Finally, in a multivariate logistic regression analysis variables of interest were assessed in fully adjusted model (Enter method). Forward and backward stepwise methods of logistic regression were then employed. To derive relative risks (RR) from odds ratios (OR), it was modified by Zhang-Yu formula.[14] To estimate population attributable risk (PAR%), [P×(RR-1)]/[1+P×(RR-1)] equation was

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employed,[15] where P is the prevalence of exposure and RR refers to fully adjusted relative risk of risk factors for incident diabetes. Data entry was carried out using EPI info™. Factors included in the analysis had less than 3% missing each; stochastic regression method was applied for imputation. Data were analysed using Stata 11.0 (Stata Corporation, College Station, TX, USA). For all analyses, statistical significance was assessed at a level of 0.05 (2-tailed).

Ethical considerations

All participants were asked to give informed consent at baseline and follow up visits according to the study protocol, approved by Isfahan Cardiovascular Research Institute Ethics Committee. Also a hardcopy of individual's own physical examination and biochemical results was handed to participants and medical consult was given where needed.

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RESULTS:

Second assessments were performed for 3283 participants in 2007. Among 6323 participants free of CVD at baseline, 1611 subjects were lost-to-follow-up, 1151did not attended, and 278 died. 303 subjects (9.2%) were also excluded as had diabetes at baseline, resulting in 2980 subjects remained consisting of 2607 normoglycemic subjects and 373 participants with prediabetes). The average age of subjects was increased from 48.8±10.2 years to 55.4±10.4 years throughout 82 (80-84) months of follow-up [median (interquartile range)].

Hypertension was slightly higher in non-examined subjects (28.5% vs. 25.4%, P=0.006). Conversely, central obesity (68.1% vs. 71.0%, P=0.015), overweight (37.8% vs. 42.4%, P<0.001) and general obesity (20.9% vs. 23.3%, P=0.024) were slightly lower at baseline. Respondent were slightly younger than non-respondent subjects (49.2±10.3 vs. 51.3±12.3 years, P<0.001). No further statistically significant differences was observed in terms of sex, smoking status, residential area, history of diabetes, hypercholesterolemia, hypertriglyceridemia, low serum HDL-C, and metabolic syndrome.

Over 7 years of follow-up, 389 (13.1%) subjects developed diabetes. New-onset diabetes was diagnosed based on isolated FPG increase, isolated 2-hour postprandial glucose (2-hPPG) increase, and combined FPG and 2-hPPG increases in 134 (34.4%), 49 (12.6%), and 65 (16.7%) subjects, respectively. Among new cases of diabetes, 92 (23.6%) subjects were identified through history of taking anti-hyperglycaemic agents (AHAs), while their plasma glucose levels were within normal levels in the follow-up visit. Forty nine (12.6%) subjects with new onset diabetes were diagnosed by physician and adopted lifestyle modification for their diabetes management. Among the new cases of diabetes, 54.2% (n=211) were aware of their disease, 32.9% (n=128) were under treatment; and 25.7% (n=100) of patients had controlled diabetes.

Furthermore, 550 (18.5%) subjects were detected to have prediabetes among which; 247 (8.3%) had IFG, 170 (5.7%) had IGT and 133 (4.5%) had combined IFG and IGT. Incidence rate of prediabetes

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was 29.1 (25.7–32.9) per 1000 person-year in men, 35.6 (31.8–39.9) per 1000 person-year in women and 32.3 (29.7–35.1) per 1000 person-year in total population. The incidence rates were almost not significantly different between men and women, while rose with aging, glucose intolerance level and increased BMI in both sex (Table 1). Exceptionally, women in 45.0-54.9 year-old age group had significantly higher incidence of diabetes in comparison with men in the same age group.

Baseline characteristics of subjects with normoglycemia are demonstrated based on their outcome at follow-up visit in three different groups in Table 2. BMI, WC, systolic blood pressure (SBP), serum triglyceride (TG), FPG, and 2-hPPG were significantly higher in those who developed diabetes compared to ones who developed prediabetes; and both groups had higher values compared to participants who sustained their normoglycemia. Weight increase was significantly lower in subjects who developed new onset diabetes in comparison to those without it, while no statistically significant difference were seen among normoglycemic and prediabetes subjects. In addition, subjects who remained normoglycemic had lower atherogenic index (TG/HDL Cholesterol) compared to prediabetes and diabetes groups, while no statistically significant difference was seen among these two groups.

Among those with prediabetes at baseline, subjects who developed diabetes had significantly higher BMI, atherogenic index, and 2-hPPG than those who retrieved from prediabetes to normoglycaemia (Table 3). FPG was significantly higher in participants who developed diabetes compared to both of the two groups who sustained prediabetes or returned to normoglycemia. Subjects with permanent prediabetes status and incident diabetes had both significantly higher TG level than those who returned to normoglycemia but no statistically significant difference was observed between them.

Considering a two level ordinal outcome in normoglycemic subjects at baseline, aging, living in rural area, unhealthy diet, overweight, obesity, ≥10% increase in WC within 7 years, high waist to hip ratio (WHR), hypertension, hypertriglyceridemia and high Atherogenic index increased the risk of

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progression to prediabetes and diabetes in the adjusted models (Table 4). In addition, higher education and ≥10% weight loss over 7 years prevented from developing prediabetes and diabetes in the adjusted models. In subjects with prediabetes at baseline, parental history of diabetes, obesity, central obesity, and hypertriglyceridemia increased risk of diabetes development. In contrast, weight gain showed a preventive effect for progression of prediabetes to diabetes. Despite lower TG, TG/HDL-C and BMI for those who retrieved to normoglycemia in bivariate analysis, no factor was found to be significantly associated with regression from prediabetes to normoglycemia.

In the full adjusted logistic regression model consisting of all factors presented in table 4, several baseline characteristics significantly contributed to the model predicting incident diabetes including parental history of diabetes [OR=1.57 (95%CI, 1.11–2.23), P=0.012, PAR%=4.83], obesity [1.49 (1.01–2.21), P=0.045, PAR%=8.54], central obesity [1.64 (1.18–2.27), P=0.003, PAR%=23.8], ≥10% weight gain [0.64 (0.42–0.98), P=0.041], hypertension [1.61 (1.24–2.09), P<0.001, PAR%=10.9], hypertriglyceridemia [1.86 (1.42–2.45), P<0.001, PAR%=30.2], isolated IGT [2.52 (1.73–3.69), P<0.001, PAR%=6.26], isolated IFG [3.30 (2.16–5.06), P<0.001, PAR%=6.06], and combined IFG/IGT [12.6(7.39–21.4), P<0.001, PAR%=9.43]. The forward and backward stepwise procedures yielded similar models with minor changes in the coefficients.

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Discussion

To our knowledge, current data is the first population-based incidence report of new onset diabetes including residents in both urban and rural areas from diverse population living in central Iran, while incidence rate of new onset prediabetes is also presented for the first time in Middle East region. In this cohort, physical activity level, dietary regimen and sleep situation has been also taken into consideration in addition to anthropometric parameters and biomarker indicators as the potential associated risk factors for glucose intolerance progression.

We reported a higher incidence rate of diabetes compared to the previous report from an urban cohort in Iran,[7] probably due to the different age structures as almost one-third of their subjects were in the 20-34 years age group with very low incidence rate of diabetes [3.1 per 1000 person-year (95% CI, 2.4–4.6)]. Furthermore, this report indicates that women aged 45-55 years, probably in perimenopause status, had significantly higher incidence rate than men in the same age group. Alterations in the pattern of sex hormones are introduced as a potential risk factor for development of diabetes in postmenopausal women.[16] The other two studies from this region reported lower incidence rates of diabetes compared to our cohort which was due to younger study population. In a Turkish prospective population-based study, total incidence of diabetes was reported 11.7 per 1000 person-year with slightly higher rates in men.[17] In another report from Iraq, 67.7% less annual incidence rate for diabetes was reported compared to our findings, but males showed higher incidence rates again.[18]

The incidence rate of diabetes in this study was also higher than European-origin populations such as Bonora et. al. study from Italy,[19] the Dutch population-based study,[20] Ely study from UK[21] and the Northern Spanish cohort of Asturias.[22] The average age of all these study population was higher than our population. All together, these findings have concordance with projections for future trend of diabetes in the world which indicates almost doubling of patients with diabetes in Eastern Mediterranean region compared to a 20% increase in European regions by 2030.[23] In addition, current

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data shows much higher incidence rate for diabetes compared to reports from an Australian cohort[24] and United States population-based studies.[25]

Blood glucose is a strong risk marker for CVDs and any level of impaired glucose regulation (IFG and/or IGT) have shown to be a leading cause of cardiovascular mortality worldwide.[1] Having said that, sizable and growing number of patients with prediabetes not only portends emerging number of patients with diabetes in near future, but also entails higher expected number of induced cardiovascular mortalities and morbidities.[26] Few studies have so far presented progression rates to impaired glucose regulation among healthy subjects,[27] whilst no data from general population in the Eastern Mediterranean region is available. In the Danish Inter-99 study,[27] progression rate from normoglycaemia to any type of impaired glucose regulation (i.e. IFG, IGT, or IFG-IGT) was considerably lower compared to our study. Remarkably high incidence rate of prediabetes in both sexes and specifically in women in our study necessitates public health interventions in order to manage current alarming trend of metabolic disorders in this region.

Unhealthy diet, overweight, obesity, central obesity, hypertension, and hypertriglyceridemia were shown to increase risk of progression to prediabetes and diabetes in our population. Obesity followed by hypertriglyceridemia was the strongest risk factor. These are in accordance with previous reports from other population-based studies for BMI,[7, 22, 28] hypertension,[7, 17, 28] hypertriglyceridemia[28] and visceral obesity.[7, 17, 20] Higher risk of diabetes in rural residential areas is explicitly in contrast with expectations and earlier published data regarding national prevalence of IFG and diabetes in Iran.[5] This could be justified by rapid life style changes, demographic transition, and urbanization in rural areas without proportionate development of health related facilities which can raise serious issues in terms of socioeconomic disparities in health. The preventive effect of advanced educational background also adds insight to the relation of socioeconomic status and diabetes. Moreover, an increased WC as low as 10% over 7 years introduced the same risk as central obesity at baseline and most other proposed risk

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factors in this study. On the other hand, 10% decrease in weight showed 33% decrease in developing prediabetes and diabetes which was independent to the baseline BMI.

Parental history of diabetes found to be risk factor for development of diabetes among prediabetes subjects. The same association was seen in normoglycemic subjects but with borderline statistical significance level. The role of parental history among prediabetes subjects has been reported previously in few other studies.[7, 24] This could be of use in screening programs to find subjects with prediabetes who are at greater risk of developing diabetes. The other associated factors with developing diabetes in subjects with prediabetes were obesity, central obesity, hypertriglyceridemia, and unexpectedly, 10% weight reduction. The detrimental effect of weight loss in this group should be due to reverse causation as subjects with prediabetes are usually advised to reduce weight.

In some other studies factors such as weight changes and obesity were introduced as effective factors for glucose intolerance improvement from prediabetes to normoglycemia.[28] Despite the fact that prediabetes subjects in our study who returned to normoglycemia had lower BMI, TG, triglycerides/HDL cholesterol ratio, FPG and 2-hPPG at baseline compared to ones who progressed to diabetes, we could not verify any statistically significant association in adjusted models.

In addition, any type of impaired glucose regulation in particular combined IFG/IGT, was found to be the most potent predictor for diabetes development like many other studies.[19, 20, 22] Results from another Iranian cohort also presented the same pattern,[7] while stronger associated role of IFG compared to IGT has already been reported in other populations with very high prevalence of diabetes.[29] However, central obesity and hypertriglyceridemia together were responsible for more than half of diabetes burden.

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Among the strengths of this study were a large population-based sample collected from both urban and rural area which covered a vast area of central Iran, and measuring a broad range of widely-accepted cardiometabolic risk factors as well as lifestyle indices by means of standard protocols. Ordered regression applied accounts for gradual ordinal nature of progression to prediabetes before diabetes. It also avoids over adjusting for intermediate outcome variable. However, the results should not be generalized without considering some limitations. Considerable rate of non-responses could be assumed as a source of bias. However, only a modest difference was observed comparing characteristics of examined and unexamined subjects. Using Heckman tow-step method, no bias was seen regarding general models predicting incident diabetes.[30] The role of some risk factors might have been over- or under-estimated in both genders by using the pooled-sex analysis. Our study might be under-powered in detecting associations in prediabetes group. Future studies should elaborate factors associated with progression to diabetes or regression to normoglycemia in this population. In addition, a generally unhealthy diet was observed to be associated with development of diabetes; however, future studies should try to scrutinise this association at level of food items in Middle-Easterners.

In conclusion, current data from this ongoing cohort demonstrated high incidence rate of diabetes in a diverse population from Eastern Mediterranean region in which, potential associated risk factors seems to be mostly modifiable. Healthcare policy makers are urged to plan and implement emergent preventive measures to control current and future burden of diabetes in the whole area.

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Table 1. Incidence of type 2 diabetes over 7 years (per 1000 person year) by age, glucose tolerance and body mass index at baseline in men and women

	Men (95% CI)	Women (95% CI)	Total (95% CI)	P-value
	16.8 (14.5-19.5)	19.9 (17.3-22.8)	18.9 (17.1–20.9)	**0.025**
Age (years)				
35-44.9	12.9 (10.0-16.5)	14.8 (11.8-18.6)	13.9 (11.7-16.4)	0.412
45-54.9	16.7 (12.5-22.4)	24.5 (19.5-30.8)	20.8 (17.4-24.9)	**0.035**
55-64.9	26.0 (19.4-34.8)	30.9 (23.1-41.4)	28.2 (22.9-34.7)	0.425
≥ 65	20.3 (13.1-31.5)	28.4 (19.1-42.4)	24.1 (17.9-32.3)	0.218
P-value*	0.001	<0.001	<0.001	
Glucose tolerance**				
NGT	12.8 (10.7-15.3)	15.5 (13.1-18.3)	14.1 (12.5-15.9)	0.119
i-IGT	41.4 (25.7-66.6)	39.6 (27.5-57.0)	40.3 (30.2-53.8)	0.916
i-IFG	46.4 (28.9-74.7)	50.1 (32.3-77.7)	48.4 (35.0-66.7)	0.726
Combined IFG/IGT	129.9 (83.0-203.7)	143.1 (99.4-205.9)	137.6 (103.7-182.5)	0.623
P-value*	<0.001	<0.001	<0.001	
BMI				
18-24.9 kg/m^2	9.9 (7.5-13.2)	14.4 (10.6-19.6)	11.5 (9.4-14.2)	0.071
25-29.9 kg/m^2	19.4 (15.7-24.1)	16.9 (13.5-21.3)	18.2 (15.5- 21.3)	0.398
≥ 30 kg/m^2	35.9 (26.6-48.6)	32.6 (26.7-39.5)	33.5 (28.5-39.4)	0.675
P-value*	<0.001	<0.001	<0.001	

CI: Confidence interval
*P-value of χ^2 test for trend of age, glucose tolerance and BMI
**NGT: Normal Glucose Tolerant; i-IFG: isolated Impaired Fasting Glucose; i-IGT: isolated Impaired Glucose Tolerance; BMI: Body Mass Index

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Table 2. Baseline characteristics of subjects with normal glucose tolerance at baseline

Baseline		**NGT (n=2607)**			
Follow-Up		**NGT (n=1893)**	**Pre-Diabetes (n=456)**	**T2D (n=258)**	**P-value**
Age (year)		47.6±9.6[bc]	50.1±11.0[a]	51.6±10.6[a]	**<0.001**
Sex (M)		908 (52.2%)	216 (47.4%)	119 (46.1%)	0.055
Leisure time PhA>1-hourper weak		443 (23.4%)	103 (22.6%)	52 (20.2%)	0.502
Sleep duration< 7-hour per night		689 (36.4%)	172 (37.7%)	110 (42.6%)	0.145
Dietary score		5.05±1.74	5.01±1.64	4.98±1.84	0.707
Smoking	Never	1455 (76.9%)	361 (79.2%)	200 (77.8%)	0.539
	Former	111 (5.9%)	31 (6.8%)	16 (6.2%)	
	Current	326 (17.2%)	64 (14.0%)	41 (16.0%)	
Parental history of T2D		193 (10.2%)	55 (12.1%)	30 (11.6%)	0.444
Body mass index (kg/m^2)		26.3±4.3[bc]	26.9±4.1[ac]	27.9±4.1[ab]	**<0.001**
Weight change from baseline (%)		2.07±9.9[c]	3.0±8.8[c]	0.42±8.9[ab]	**0.003**
Waist circumference (cm)		93.4±11.8[bc]	95.2±11.7[ac]	97.8±11.9[ab]	**<0.001**
	Change from baseline (%)	0.42±12.9	0.97±13.6	0.23±14.5	0.682
Systolic blood pressure (mmHg)		117±18.4[bc]	122±19.8[ac]	127±21.2[ab]	**<0.001**
Diastolic blood pressure (mmHg)		76.5±11.1[bc]	78.3±10.5[ac]	81.0±11.6[ab]	**<0.001**
Triglycerides (mg/dl)		155±94.7[bc]	172±105[ac]	197±102[ab]	**<0.001**
LDL cholesterol (mg/dl)		126±42.2	126±44.1	131±44.3	0.247
HDL cholesterol (mg/dl)		46.4±10.2	47.2±10.3	48.0±10.5	**0.038**[†]
Triglycerides/HDL cholesterol ratio		3.41±2.4[bc]	3.73±2.6[a]	4.20±2.3[a]	**<0.001**
Total/HDL cholesterol ratio		4.6±1.3	4.7±1.4	4.8±1.27	0.201
C-reactive protein (mg/l)		3.3±1.5	3.2±1.5	3.26±1.6	0.963
Fasting plasma glucose (mg/dl)		77.7±8.7[bc]	79.0±8.5[ac]	81.1±9.5[ab]	**<0.001**
2-hour post prandial glucose (mg/dl)		90.1±17.0[bc]	94.9±19.2[ac]	99.1±19.6[ab]	**<0.001**

Data are expressed as mean ± SD for continuous variables and number (percentage) for categorical variables
NGT: Normal glucose tolerant; Pre-diabetes: including impaired fasting glucose and/or impaired glucose tolerance; T2D: Type 2 diabetes mellitus; PhA: Physical activity
a, b, c: P<0.05 based on Bonferroni post hoc test for pairwise comparisons (a, b, and c indicate first, second, and third groups, respectively)
† A borderline significant difference was found between a and c (P = 0.063)

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Table 3. Baseline characteristics of subjects with pre-diabetes at baseline

Baseline		**Pre-Diabetes (n=373)**			
Follow-Up		**NGT (n=148)**	**Pre-Diabetes (n=94)**	**T2D (n=131)**	**P-value**
Age (year)		51.6±11.8	50.9±11.5	51.3±9.8	0.894
Sex (M)		65 (43.9%)	36 (38.3%)	53 (40.5%)	0.668
Leisure time PhA >1-hour per weak		27 (18.2%)	19 (20.2%)	28 (21.4%)	0.803
Sleep duration< 7-hour per night		63 (42.6%)	33 (35.1%)	46 (35.1%)	0.349
Dietary score		5.18±1.74	4.98±1.51	5.21±1.73	0.560
Smoking	Never	124 (83.8%)	78 (83.0%)	113 (86.3%)	0.361
	Former	4 (2.7%)	7 (7.4%)	6 (4.6%)	
	Current	20 (13.5%)	9 (9.6%)	12 (9.2%)	
Parental history of T2D		13 (8.8%)[c]	9 (9.6%)	26 (19.8%)[a]	**0.012**
Body mass index (kg/m^2)		27.1±4.37[c]	28.4±4.49	29.4±4.53[a]	**<0.001**
Weight change from baseline (%)		0.98±11.4	0.67±8.39	-1.77±9.58	0.055
Waist circumference (cm)		94.3±12.7	98.2±12.1	101±11.5	0.061
Change from baseline (%)		2.04±17.3	-0.04±10.9	-1.59±11.3	0.092
Systolic blood pressure (mmHg)		123±20. 9	127±24.4	127±21.2	0.151
Diastolic blood pressure (mmHg)		78.4±10.3	80.4±11.2	80.9±11.1	0.132
Triglycerides (mg/dl)		173±91.2[bc]	201±1.64[a]	208±1.57[a]	**0.003**
LDL cholesterol (mg/dl)		135±39.5	142±49.6	138±46.6	0.427
HDL cholesterol (mg/dl)		46.8±10.3	46.8±10.3	47.6±11.3	0.804
Triglycerides/HDL cholesterol ratio		3.78±2.43[c]	4.39±1.78	4.48±1.70[a]	**0.011**
Total/HDL cholesterol ratio		4.85±1.31	5.19±1.59	5.09±1.51	0.176
C-reactive protein (mg/l)		3.14±1.48	3.09±1.41	3.49±1.94	0.220
Fasting plasma glucose (mg/dl)		94.1±13.6[c]	93.2±13.4[c]	102±12.0[ab]	**<0.001**
2-hour post prandial glucose (mg/dl)		140±28.7[c]	145±27.1	151±27.0[a]	**0.003**

Data are expressed as mean ± SD for continuous variables and number (percentage) for categorical variables
NGT: Normal glucose tolerant; Pre-diabetes: including impaired fasting glucose and/or impaired glucose tolerance; T2D: Type 2 diabetes mellitus; PhA: Physical activity
a, b, c: P<0.05 based on Bonferroni post hoc test for pairwise comparisons (a, b, and c indicate first, second, and third groups, respectively)

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Table 4. Factors associated with development of prediabetes and type 2 diabetes in adjusted models

Baseline		**NGT**		**Pre-Diabetes**			
Follow-Up		**→Pre-diabetes→T2D**		**NGT ←**		**→T2D**	
		OR (95% CI)*	**P-value**	**OR (95% CI)***	**P-value**	**OR (95% CI)***	**P-value**
Age (year)		1.03 (1.02- 1.04)	**<0.001**	1.01 (0.99-1.03)	0.428	1.01 (0.99-1.03)	0.356
Sex (M)		0.90 (0.75- 1.08)	0.251	1.21 (0.74-1.98)	0.442	1.12 (0.71-1.75)	0.633
Education	Illiterate/Basic	1		1		1	
	High school/College	0.76 (0.61-0.95)	**0.018**	0.92 (0.51-1.66)	0.794	1.63 (0.95-2.77)	0.074
	University	0.72 (0.56-0.92)	**0.009**	0.63 (0.32-1.25)	0.190	0.99 (0.52-1.90)	0.980
Population type (Rural)		1.28 (1.05- 1.55)	**0.014**	0.95 (0.56-1.60)	0.844	0.77 (0.47-1.25)	0.293
Leisure time PhA> 1 hour per weak		0.90 (0.73 -1.11)	0.309	0.94 (0.51-1.73)	0.855	1.16 (0.66-2.03)	0.609
Dietary score Q1		1.32 (1.03- 1.69)	**0.025**	0.78 (0.59-1.05)	0.102	0.71 (0.36-1.40)	0.326
	Q2	1.20 (0.91- 1.58)	0.191	0.83 (0.60-1.15)	0.262	0.77 (0.39-1.48)	0.432
	Q3	1.11(0.84- 1.47)	0.447	0.89 (0.64-1.24)	0.493	0.94 (0.53-1.70)	0.858
	Q4	1		1		1	
Smoking Current		1.09 (0.82- 1.41)	0.580	1.06 (0.77-1.45)	0.696	1.28 (0.86-1.90)	0.227
	Former	0.85 (0.56- 1.32)	0.511	1.03 (0.59-1.32)	0.479	1.03 (0.58-1.82)	0.907
	Never	1		1			
Sleep duration ≥ 7 hours/night		1.07 (0.89- 1.27)	0.473	1.23 (0.75-2.01)	0.405	0.81 (0.51-1.27)	0.356
Parental history of T2D		1.30 (0.98- 1.70)	0.063	1 .74 (0.78-3.87)	0.175	2.26 (1.18-4.33)	**0.014**
Body mass index							
	25-29.9 kg/m^2	1.45 (1.18- 1.78)	**<0.001**	0.58 (0.31-1.09)	0.090	1.35 (0.74-2.46)	0.320
	≥ 30 kg/m^2	1.97 (1.55-2.52)	**<0.001**	0.78 (0.39-1.55)	0.483	2.59 (1.41-4.77)	**0.002**
Weight change							
	Reduced (≥10%)	0.67 (0.47-0.96)	**0.030**	1.94 (0.77-4.88)	0.160	2.93 (1.45-5.91)	**0.003**
	Unchanged (<10%)	1					
	Increased (≥10%)	0.90 (0.70-1.18)	0.460	0.74 (0.34-1.63)	0.462	1.04 (0.46-2.34)	0.924
Waist circumference ≥93/94 cm†		1.13 (0.92-1.39)	0.255	0.91 (0.50-1.69)	0.777	2.03 (1.11-3.74)	**0.022**
Waist circumference change							
	Reduced (≥10%)	1.03 (0.78- 1.37)	0.829	1.25 (0.65-2.39)	0.498	1.06 (0.60-1.80)	0.833
	Unchanged (<10%)	1		1		1	
	Increased (≥10%)	1.37 (1.06-1.78)	**0.017**	1.12 (0.54-2.34)	0.754	0.90 (0.45-1.80)	0.771
Waist-to-hip ratio >0.80/0.95		1.35 (1.01-1.66)	**0.004**	1.14 (0.55-2.37)	0.717	1.40 (0.70-2.78)	0.341
Hypertension		1.35 (1.10-1.66)§	**0.004**	0.77 (0.43-1.39)	0.373	1.55 (0.93-2.60)	0.095
Triglycerides ≥150 mg/dl		1.50(1.25-1.81)†	**<0.001**	0.92 (0.53-1.59)	0.762	1.94 (1.14-3.32)	**0.015**
LDL cholesterol ≥130 mg/dl		0.97(0.81-1.16)	0.741	0.94 (0.58-1.54)	0.832	1.15 (0.74-1.80)	0.530
HDL cholesterol ≤40/50§ mg/dl		1.00 (0.84-1.20)	0.979	1.00 (0.61-1.64)	0.997	0.82 (0.52-1.29)	0.390
Triglycerides/HDL-C ratio >4		1.45 (1.22-1.74)	**<0.001**	0.71 (0.44-1.15)	0.168	1.42 (0.91-2.21)	0.123
Total/HDL cholesterol ratio ≥5		0.98 (0.82-1.17)	0.823	0.76 (0.48-1.23)	0.270	1.06 (0.68-1.64)	0.805
C-reactive protein (mg/l) >3		0.94 (0.76-1.17)	0.603	0.84 (0.48-1.48)	0.558	1.36 (0.80-2.33)	0.258
Metabolic syndrome		1.27 (1.05-1.54)	0.015	1.02 (0.69-2.13)	0.549	1.61 (0.93-2.77)	0.088

* Odds ratio adjusted for age and sex, parental history of T2D and body mass index classification
NGT: Normal glucose tolerant; Pre-Diabetes: IFG and/or IGT; T2D: Type 2 diabetes mellitus; PhA: Physical activity
§ P-value by Brant test: 0.004; separated OR was 1.26 (1.02-1.56, P=0.029) for Pre-Diabetes and 1.90 (1.42-2.54, P<0.001).
† P-value by Brant test: 0.013; separated OR was 1.44 (1.20-1.74, P<0.001) for Pre-Diabetes and 2.07 (1.53-2.81, P<0.001).

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